

EE 102: Signal Processing and Linear Systems

Problem Set #6: Fourier Series

Name: _____

Submission Date: _____

Problem 1 Consider the RC circuit from Pre-requisite #3, with input voltage $x(t)$ and output voltage $y(t)$ measured across the capacitor. Use $R = 330\ \Omega$ and $C = 1000\ \mu\text{F}$, so $RC = 0.33\text{ s}$. Its impulse response is

$$h(t) = \frac{1}{RC} e^{-t/(RC)} u(t).$$

For each of the input signals below, find the Fourier series representation of the **periodic steady-state output** $y(t)$.

(a) [20 points] A train of impulses:

$$x(t) = \sum_{n=-\infty}^{\infty} \delta(t - n)$$

Hint: You can find the Fourier series representation of the input signal and then use the property of LTI systems that relates the Fourier series coefficients of the input and output signals.

(b) [20 points] A square wave of amplitude 1 sketched below in Figure 1.

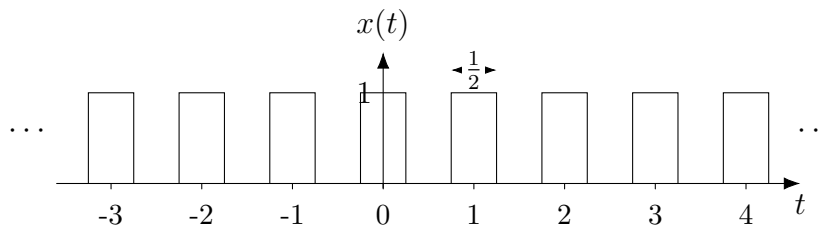


Figure 1: A square wave of amplitude 1, period $T = 1$, and duty cycle 50%.

Hint: Over one period, $-\frac{1}{2} \leq t < \frac{1}{2}$, the signal is

$$x(t) = \begin{cases} 1, & -\frac{1}{4} \leq t \leq \frac{1}{4}, \\ 0, & \text{otherwise within this period.} \end{cases}$$

(c) [20 points] The input $x(t)$ in part (b) models a PWM signal with a fixed duty cycle. Double its switching frequency while keeping the amplitude and duty cycle unchanged:

$$x_{\text{new}}(t) = x(2t).$$

Sketch this new input and state its fundamental period. Use the Fourier series coefficients from part (b), the time-scaling property, and the LTI system's response at the new harmonic frequencies to find the Fourier series representation of the new output.

Hint: Time scaling changes the harmonic frequencies while retaining the input coefficient sequence. Evaluate the system response at those new frequencies; the new output is not generally a time-scaled copy of the output in part (b). You do not need to recompute the input coefficients from scratch.

(d) [10 points] Write Python/MATLAB code to implement the signals in (b) and (c). Compute the outputs and their Fourier series representations, and plot the magnitude spectra of the outputs. Validate your analytical results against simulations in periodic steady state, after startup transients have decayed. State your sampling interval and the number of harmonics used.

Problem 2 Compute the complex Fourier series representation of each signal below. For each signal $f(t)$, let c_k be its Fourier series coefficients and $\omega_0 = 2\pi/T_0$. Define the truncated representation by

$$f_N(t) = \sum_{k=-N}^N c_k e^{jk\omega_0 t}.$$

Thus, retain the DC term and the first N positive and negative harmonics. For $N = 1, 2, 3, 4$, compute the error energy **over one period**:

$$E_N = \int_0^{T_0} |f(t) - f_N(t)|^2 dt.$$

You may use computer programs for computations. Show how you obtain the coefficients and the error energy.

(a) [15 points] A sawtooth signal with fundamental period $T_0 = 1$. Figure 2(a) shows one period:

$$x(t) = t, \quad 0 \leq t < 1, \quad x(t+1) = x(t).$$

Extend this definition periodically to all positive and negative time.

Hint: Evaluate the coefficient integral over $0 \leq t < 1$. Integration by parts can be useful for the nonzero harmonics.

(b) [15 points] A rectangular signal with fundamental period $T_0 = 2$, shown in Figure 2(b). Over one period,

$$y(t) = \begin{cases} 1, & 0 \leq t < 1, \\ 0, & 1 \leq t < 2, \end{cases} \quad y(t+2) = y(t).$$

The signal repeats periodically for all positive and negative time.

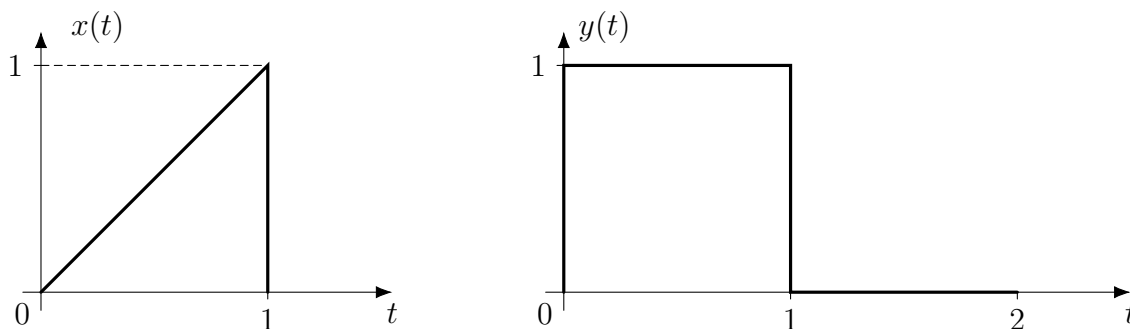


Figure 2: One period of (a) the sawtooth signal $x(t)$ and (b) the rectangular signal $y(t)$.